

Flexural Vane Pump

Joyce Noh

Department of Mechanical Engineering
Massachusetts Institute of Technology
Cambridge, MA, USA
jnoh@mit.edu

Abstract—In this study, a standard vane pump is redesigned using flexures to potentially solve the issues of reducing friction between the vanes and the stator walls while constantly maintaining contact. The vane pumps in study were to be used in a desalination plant system.

I. INTRODUCTION

A. Rotary Vane Pump

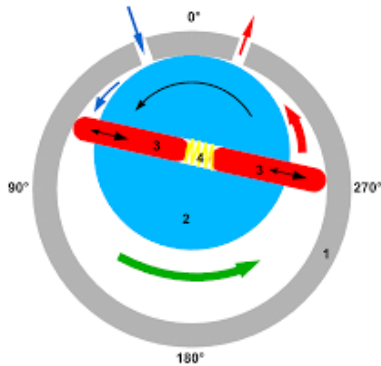


Figure 1: Standard rotary vane pump

A rotary vane pump is a positive-displacement pump that uses vanes mounted in a rotating rotor to move the fluid. The vanes move in and out of a channel or cavity in the rotor while maintaining contact with the surrounding stator walls. This contact is crucial as any leakage of the fluid would undermine the pump's functionality. At the same time, the pump runs more efficiently when the friction between the vane and the stator walls is kept to a minimum. Centrifugal or high pressure vanes are some of the commonly used designs. However, these designs can cause jamming because of its geometry and loads.

B. Approach

The main goal of this study is to design and compare different flexural solutions that could be used to allow the vane to maintain contact with the stator wall while minimizing friction between the vane and the stator wall.

The two types of flexures explored are flexure strips and flexure coils. Different configurations of each are designed to achieve the linear motion of the vane within the

rotor. After selecting the best idea for each type of flexure, the designs will be modeled and prototyped using 3D printing. Each design will then be evaluated against certain functional requirements.

II. DESIGN SOLUTIONS

A. Design 1: One-Chamber Flexure Strip

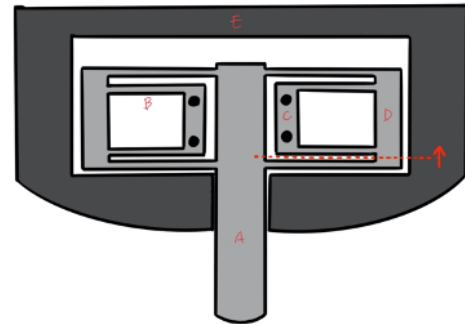


Figure 2: One-chamber design sketch. Top view.

This design uses two extra pairs of flexure strips (B) that are all attached by a freely moving piece (D). The two extra pairs are used to connect to stationary pieces (C) that are firmly attached to the rotor (E). This allows for a more stable linear motion and minimal potential for racking from the vane (A). Instead of two chambers, this design only has one chamber to house the flexures.

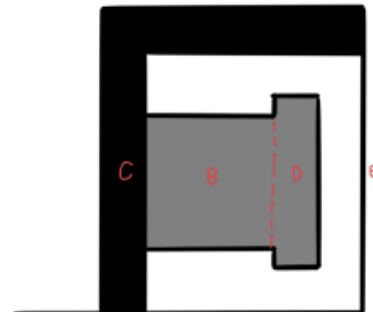


Figure 3: Cross-sectional view sketch of one-chamber design.

This is a cross-sectional view of the design along the shown dotted line and in the direction of the red arrow. As

mentioned above, part (C) is stationary and attached firmly to the rotor (E) while part (D) is free to move. They are connected by the flexure pairs (B). In order to achieve this, (C) is constrained to the rotor at two points at the bottom and top in order to prevent instability and deflection. The shown flexure (B) is the inner pair of flexures.

Overall, this solution creates much more robust linear motion, little potential for racking, and larger range of motion. However, it is a considerably complex design and would, therefore, be more difficult to manufacture, especially as one assembly or print.

B. Design 2: Two-Chamber Flexure Strip

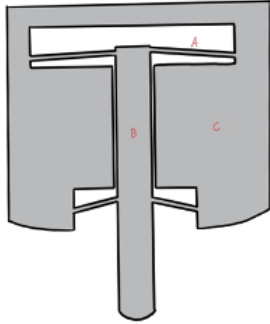


Figure 4: Two-chamber design sketch. Top view.

This design uses thin flexural strips (A) to allow for motion. By constraining the vane (B) at both ends, the vane can move along the channel linearly. The two chambers separated by the middle channel allows for the flexure pairs to be as far away from each other as possible to reduce the potential of racking. In terms of manufacturing, it would be best to print as one assembly with the rotor (C) as the flexures need to be precise in length and not allow for slipping.

This is a much simpler design that still creates linear motion. However, the flexure strips in this design are subject to both bending as well as stretching and have less range of motion than the first design.

C. Design 3: Robust Flexure Strip Design

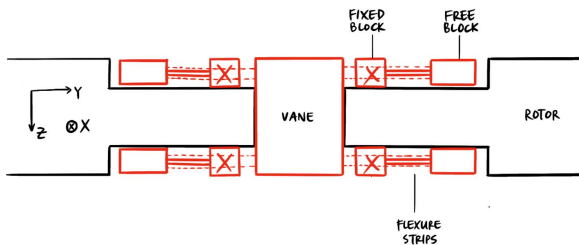


Figure 5: Front view of vane, attached flexure strips, and rotor.

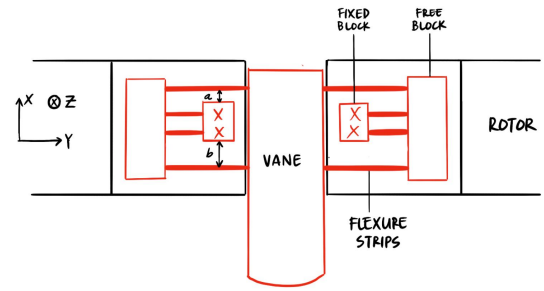


Figure 6: Top view of vane, attached flexure strips, and rotor. Dimensions a and b determine the range of motion of the vane along the x-axis.

This design uses thin flexure strips to allow for motion. The linear motion of the vane is achieved in two ways. First, by constraining the vane in two main areas the vane can move along the channel without significant racking. Second, one assembly of the flexure strips consists of one fixed block and one free block as shown in the figures above. There are two different flexure lengths: the length of the flexure strips between the fixed and free blocks and the length of the flexure strips between the free block and the vane. This allows for the vane to achieve a significant range of linear motion within the channel. The dimensions a and b , as shown in Figure 2, will determine the range of motion along the x-axis.

There are two chambers, as shown in Figure 1, separated by an extension of the rotor. The extension of the rotor serves as the anchor for the fixed blocks of the four flexure strip assemblies.

This design will be manufactured using 3D printing solutions in two ways. First, the red portion (vane, flexure strips, fixed block, and free block) and the black portion (rotor) in the figures above will be printed as two separate pieces. Then the red portion will be assembled to the rotor by attaching the fixed blocks. Another method of manufacturing is to print the entire design as one assembly. This would eliminate the need for further assembly processes.

This solution allows the flexure strips to withstand less stretching as they bend while still creating linear motion and a significant range of motion depending on dimensions a and b , as mentioned earlier. However, the flexure strips of equal length must be very precise adding to the overall complexity of the design's manufacturability and assembly.

D. Design 4: Flexural Coils

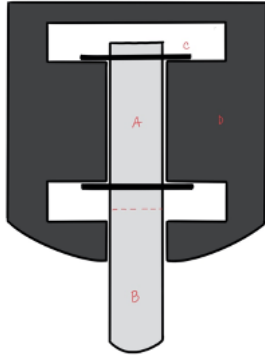


Figure 7: Flexural coil design sketch. Top View.

This flexure vane design uses coil flexures instead of strips. These coils (C) are attached to the vane at two ends, both contained in two separate chambers. The two separate chambers allow the coils to be as far from each other as possible to minimize the potential of racking and to minimize leakage of the fluid. These coils allow the vane (A) to travel through the channel linearly. Furthermore, being constrained by the two coils, the vane will be contact-free from the surround channel wall of the rotor (D). This minimizes the friction between the channel wall and the vane to just the area surrounding portion B of the vane.

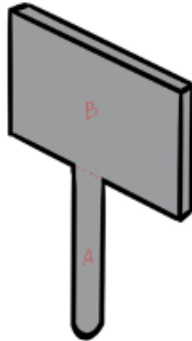


Figure 8: Vane design sketch.

This solution calls for a redesigned vane as well. Part (B) of the vane is much like any typical vane in a vane pump. Part (A) is the part constrained by the concentric coil flexures.

This solution creates stable linear motion, has minimal potential for racking, allows a large range of motion, and keeps the vane in contact with the stator walls without applying too much unnecessary force. The potential risks of this design is that it is moderately complex in design and manufacturability and would most likely not be built or printed as one assembly.

III. FUNCTIONAL REQUIREMENTS

When selecting the final design solution, a few functional requirements were considered:

1. It must allow the vane to maintain contact with the stator wall.

2. It must minimize friction both between the vane and the channel as well as between the vane and the stator wall.
3. Failure potential must be considered in terms of the stresses that the design experiences.
4. The solution must be relatively simple in design and manufacturability.

A. Selection

After reviewing the designs' functional requirements, the designs chosen are Design 3 and 4.

TABLE 1: DESIGN SELECTION - PUGH CHART

	1	2	3	4	Total
Design 1	0	0	0	0	0
Design 2	0	+	-	-	-
Design 3	0	+	+	-	+
Design 4	+	+	-	0	+

IV. FLEXURE STRIP: PROTOTYPING

A. Modeling and Review

The following images show the CAD models created on Autodesk Fusion 360.

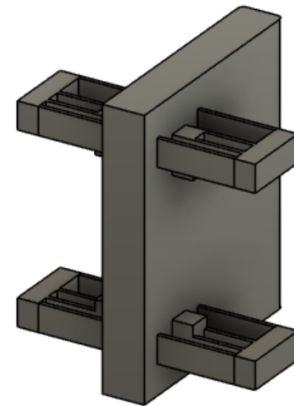


Figure 9: Vane CAD with attached flexure strips.

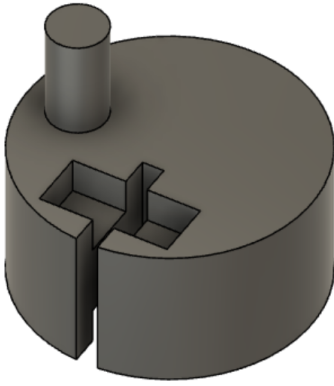


Figure 10: Rotor CAD.

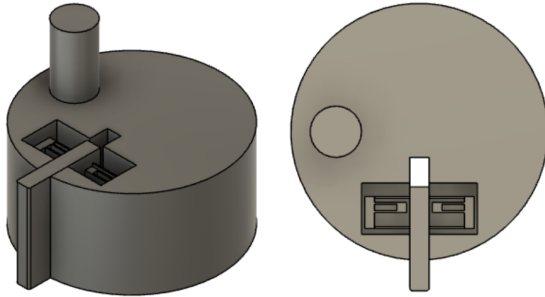


Figure 11: Assembly CAD of vane and rotor.

Each part was then 3D printed using the Stratasys Fortus 400MC Printer which uses Fused Deposition Modeling (FDM) technology.



Figure 12: 3D-printed vane with attached flexure strips..

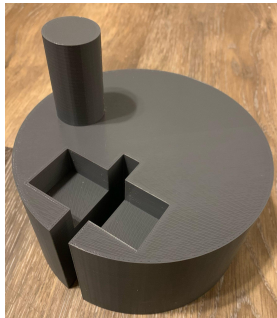


Figure 13: 3D-printed rotor.

Considering the functional requirements, there were two main concerns with this design. First, since the flexure strips (A) are made from 3D-printed plastic, it does not have the same flexible properties as thin metal flexure strips. Under very little strain, the plastic flexure strips break, limiting the vane's range of motion (B).

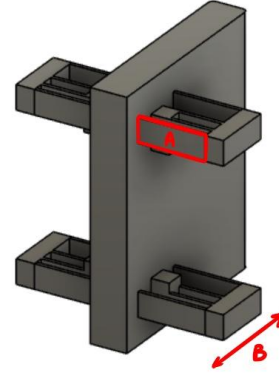


Figure 14: Vane and flexure strip assembly CAD. (A) and (B) denote the areas with the main issues with the initial vane assembly design.

Second, when the vane is printed as one assembly with the main body and the flexure groups, it cannot be placed into the rotor. The four flexure groups are divided into two main chambers: top and bottom as shown (C). Since these chambers are cut within the rotor and the wall opening of the rotor only has a channel for the vane to travel through (D), the vane assembly cannot be fixed in the rotor.

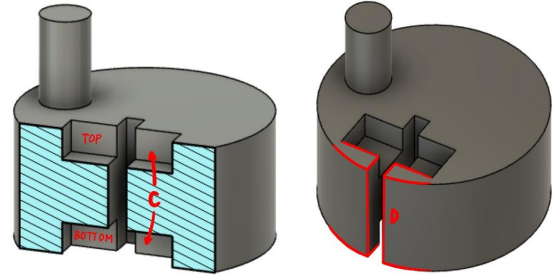


Figure 15: Rotor CAD. (C) and (D) denote the main issues with the initial rotor design.

B. Future Iterations

Future iterations would consider using different materials for the flexure strips such as thin sheets of metal and printing the vane as separate parts that can be fitted into the rotor. The vane can be separated into three pieces: the top flexure strip pair, the middle connector, and the bottom flexure strip pair.

V. FLEXURE COIL: PROTOTYPING

A. Modeling and Review

The flexural coil design was modeled through CAD and 3D printed using the same method as the flexure strip design.

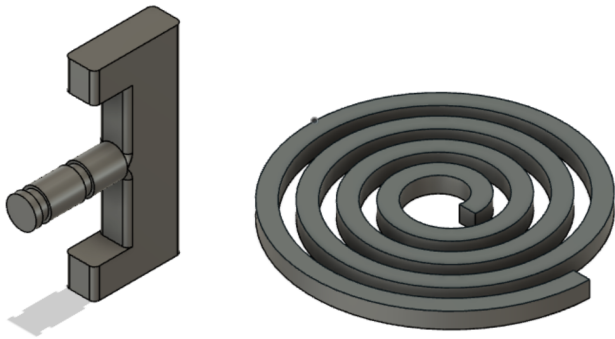


Figure 16: Vane CAD for coil design solution and flexural coils.

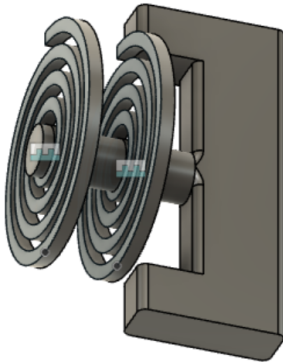


Figure 17: Assembly CAD of vane and two coils.

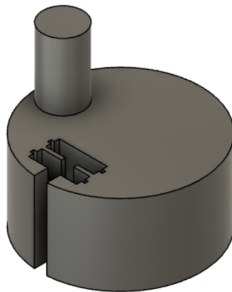


Figure 18: Rotor CAD.

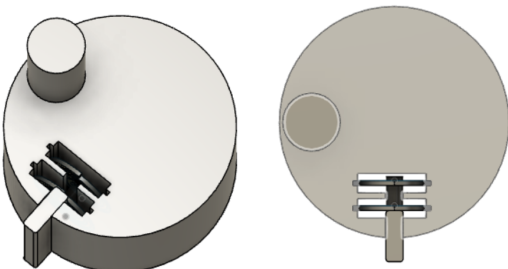


Figure 19: Total assembly of vane, coils, and rotor.

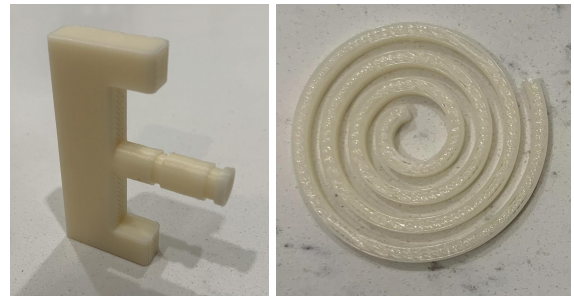


Figure 20: 3D-printed vane and coil.

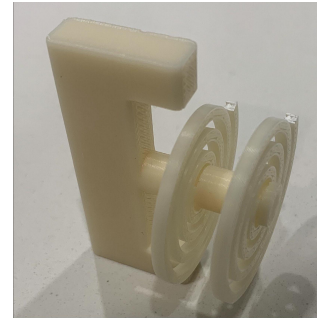


Figure 21: 3D-printed assembly of vane and two coils.

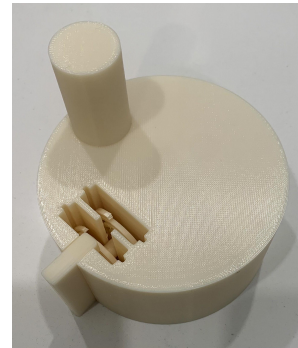


Figure 22: Vane and coil assembly fitted into rotor.



Figure 23: Total assembly of vane, coils, rotor, and stator.

After reviewing the printed prototype, there were a few notable considerations. First, the complete assembly of the vane and two coils was not expected to print well compared to the assembly of three separately printed parts. However, the complete assembly proved to be stronger and closer to the desired dimensions. Furthermore, the coils were

expected to be much stiffer. However, the resulting low stiffness of the coils allowed for a greater range of motion for the entire vane assembly. The main risks were seen in two areas. First, the tight slot dimensions for the coils did not allow the coils to seamlessly slide in (A). Second, the length of the back chamber was too short as the vane was blocked by the back wall when fully compressed (B).

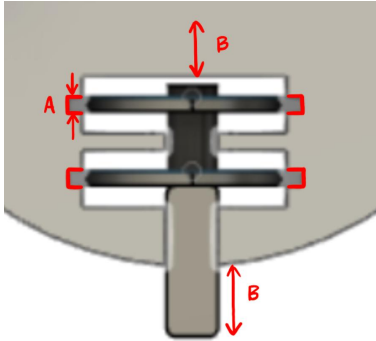


Figure 24: Vane and coil flexure assembly inside the rotor. (A) and (B) denote the main issues with this design.

B. Technical Review II

From the previous design, there are four main differences. (A) First, the wall separating the two chambers has been removed in order for the coils to have a greater range of motion. (B) The back wall has been extended in order to remove the obstacle in the vane's path. (C) The dimension shown at the front of the vane has also been reduced so that the vane does not collide with the coil flexure as it is compressed into the rotor. (D) Lastly, the width of the slots in the rotor for the coil flexures has been increased to allow the coils to have a close but smooth fit.

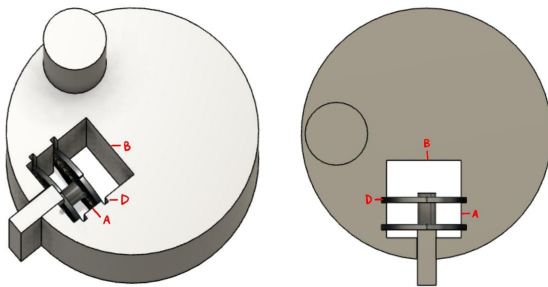


Figure 25: Angled and top view of the new assembly CAD.

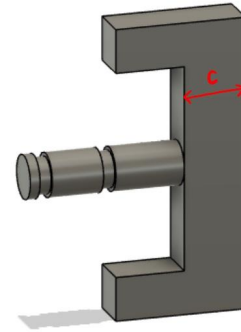


Figure 26: Updated vane CAD.

C. Future Iterations

The final design of the coil flexure solution proved to meet all the critical functional requirements except in the area of the design's complexity. While the design maintains contact between the vane and the outer stator without applying too much force to counter the contact force, performs linear motion with little room for racking, and minimizes overall friction, it takes up a significant portion of the rotor. This limits how many vanes can be fitted into the rotor or requires a very large rotor which would only be used in quite sizable pumps. A future iteration of this design would maintain its mechanical features while reducing the footprint of the cavity in the rotor itself.

ACKNOWLEDGMENTS

I would like to thank the MIT GEAR Lab for funding and supporting this project by lending its resources. I would also like to extend a special thanks to Sahil Shah for supervising by providing guidance and knowledge throughout the course of the project.

REFERENCES

- [1] Wikimedia Foundation. (2022, March 7). Rotary Vane Pump. Wikipedia. Retrieved March 15, 2022, from https://en.wikipedia.org/wiki/Rotary_vane_pump